

Region Segmentation based on Fluorescent Imaging

Anonymous CVPR submission

Paper ID *****

Abstract

001 *Hospital-associated saliva deposition can seed high-risk*
002 *pathogens across shared surfaces. This work studies un-*
003 *supervised segmentation of saliva deposits illuminated un-*
004 *der UV-C, where only ten color images are available and*
005 *ground truth masks only cover four centralized patches per*
006 *frame. We explore a hierarchy of methods: a MATLAB-*
007 *based luminance/thresholding pipeline (Method A), an ex-*
008 *ploratory k-means segmentation (Method C), and our prob-*
009 *abilistic Gaussian mixture model (Method D) that reuses*
010 *k-means centers, runs ten EM initializations per image, and*
011 *selects the most likely fit by pseudo-log-likelihood. The re-*
012 *sulting GMM probability maps are binarized with morpho-*
013 *logical cleanup and evaluated on accuracy, precision, re-*
014 *call, and Dice; the $K=4$ model achieves the highest Dice*
015 *while maintaining recall above 0.96. Our comparison high-*
016 *lights how converting hard clusters into likelihood-ranked*
017 *mixtures improves recall while still recovering reasonable*
018 *precision, providing a practical unsupervised pipeline for*
019 *low-data airborne contamination imagery.*

020 1. Introduction

021 1.1. Problem Definition

022 Hospital-acquired infections (HAIs) pose a recurring threat
023 to healthcare environments, increasing patient risk, compli-
024 cating workflows, and incurring higher operational costs.
025 HAIs spread by direct contact, droplets, aerosols, and con-
026 taminated surfaces. The problem is further harmful in
027 busy clinical areas, where equipment is moved frequently,
028 and patients are vulnerable. Major reviews and guidelines
029 identify environmental reservoirs, device-related contami-
030 nation, and workflow pressures as the primary factors con-
031 tributing to disease propagation in hospitals, suggesting that
032 broad infection-control policies alone are insufficient. Ad-
033 ditionally, knowing where contamination actually accumu-
034 lates is crucial, allowing for focused and effective cleaning
035 [1, 8, 9, 12, 19].

036 Saliva is a common target for efforts aimed at reduc-

ing HAIs because it carries many pathogens and is easily
spread during talking, coughing, and routine procedures.
Once saliva lands on instruments or surfaces, its appear-
ance changes as it dries or interacts with different materials,
making it difficult to spot under normal lighting. Studies
using fluorescence reveal clear differences between moist
and dried saliva on various surfaces, indicating that opti-
cal imaging can detect deposits that would otherwise be
missed [11, 20]. Other reviews and studies that treat saliva
as a diagnostic and surveillance fluid also highlight its role
in pathogen carriage and transmission, reinforcing the idea
that detecting saliva specifically is a useful way to reduce
HAIs [3, 10, 18].

In practice, hospitals employ a layered approach to pre-
vention, utilizing masks, face shields, and routine cleaning
and sterilization. Masks and shields reduce the spread of
droplets but do not eliminate surface deposition. Experi-
ments that use fluorescent surrogates often reveal contam-
ination patterns invisible to the naked eye, demonstrating
that barriers and cleaning must work together [17]. Mask
effectiveness depends on fit, material, and how long it is
worn. Leaks and saturation can still let droplets escape
and contaminate nearby surfaces [2]. Face shields block
splashes and large droplets but do not prevent peripheral
deposition, and the shield surfaces themselves can become
contaminated. Therefore, they are most effective when used
in conjunction with masks [2, 17]. Routine disinfection
lowers surface bioburden, but actual coverage and contact
time vary in real settings. Laser-induced fluorescence (LIF)
can map where droplets land and persist, providing useful
feedback to improve cleaning patterns and identify missed
spots [20].

Computer vision can make these imaging findings ac-
tionable. Segmentation algorithms can produce binary
masks that mark saliva or fluorescent surrogates in images.
These masks can guide automated or semi-automated clean-
ing tools to focus on contaminated regions. Fluorescence
imaging and high-speed visualization provide the contrast
and ground truth necessary to train and validate these al-
gorithms across various surface types and moisture states,
helping bridge the gap between lab measurements and real-

078 world cleaning workflows [2, 11, 17, 20].

079 1.2. Literature Review

080 Medical image segmentation is the area of medical tech-
081 nology that most resembles the creation of binary masks
082 for saliva recognition. This process labels every pixel (or
083 voxel) in an image to isolate organs, lesions, fluids, usu-
084 ally by difference in contrast. For contamination detection
085 this usually boils down to a binary mask (contaminant vs.
086 background), but the goals are the same as other clinical
087 tasks: get accurate boundaries, be robust to different imag-
088 ing conditions, and produce outputs that can drive decisions
089 or automated actions. Standard overlap metrics such as the
090 Dice Coefficient and Intersection over Union (IoU) are used
091 to judge how well a model matches expert masks, called
092 Ground Truths (GT). These metrics guide model selection
093 and tuning [15].

094 The methods for segmentation range from image-
095 processing techniques to deep neural networks. Basic steps,
096 such as contrast enhancement, thresholding, edge filters
097 (Sobel/Laplace), and morphological cleanup, are still used
098 for boosting visibility of small or low-contrast deposits, as
099 well as for preprocessing images before applying the model
100 [4, 15]. On the learning side, U-Net and its many variants
101 are the baseline because they combine multi-scale context
102 with precise localization. More recent work adds attention
103 blocks or mixes CNNs with transformers to capture longer-
104 range context and improve small-object detection [6, 14].
105 There’s also a growing push to use synthetic data and gener-
106 ative models to expand training sets, and public codebases
107 for droplet tasks (e.g., Droplet-ML) make it easier to repro-
108 duce experiments and try augmentation strategies [13, 16].

109 The main challenges for saliva/droplet segmentation are
110 familiar in the computer vision field. Getting enough la-
111 beled masks is expensive, droplets are tiny so class imbal-
112 ance is severe, and appearance changes a lot with surface
113 type, lighting, and drying state. That means the necessity
114 of specialized datasets, multi-scale supervision, and care-
115 ful preprocessing or domain-adaptation to generalize from
116 lab images to real clinical scenes [6, 14]. Practically speak-
117 ing, the best routes adopt a hybrid approach, using classical
118 enhancement to make deposits visible [15], training robust
119 segmentation networks (with attention to small-object han-
120 dling) [6], and relying on synthetic or open datasets to fill
121 gaps [13, 16].

122 1.3. Contributions

123 In this work we deliver:

- 124 • A structured comparison of the existing MAT-
125 LAB/thresholding pipeline (Method A), the exploratory
126 k-means clustering pipeline (Method C), and the prob-
127 abilistic Gaussian mixture model pipeline (Method
128 D).

- A GMM workflow that seeds EM with k-means centers, runs ten initializations per image, selects the best solution via pseudo-log-likelihood, and thresholds the resulting probability map before morphological cleanup.
- Empirical evidence that the $K = 4$ GMM increases Dice while keeping recall above 0.96, illustrating how likelihood-aware model selection improves the unsupervised segmentation of saliva under UV-C.

2. Dataset and Methods

2.1. Dataset description

The dataset contains ten UV-C images of saliva deposits on five substrates (aluminum, steel, white wood, PVC, and wood) with both wet and dry patches. Each frame is accompanied by a binary mask that highlights four roughly centralized saliva patches. We treat the first eight images as the tuning set and reserve the final two images for blind testing; the masks only cover the salient regions, so every method must operate in an unsupervised fashion while relying heavily on color, contrast, and spatial locality.

- **Imaging:** High-contrast UV-C photography with fluorescent saliva to accentuate deposits across heterogeneous materials.
- **Ground truth:** One binary mask per image that marks four foreground regions, which guide plausibility filters and evaluation.
- **Split:** Images 1-8 serve for parameter tuning, and 9-10 are used to report unseen performance.
- **Challenge:** The small dataset prevents supervised training, so all methods rely on heuristics shaped by saliva brightness and location.

2.2. Method A: MATLAB thresholding pipeline

Method A is a lightweight MATLAB pipeline that maximizes recall through contrast enhancement and biased thresholding. We convert RGB frames to luminance using ITU-R BT.601 weights, apply CLAHE to normalize local contrast, and tune a bias for Otsu’s threshold so that the resulting binary mask includes more saliva when the default threshold underestimates the foreground.

2.2.1. Pre-processing

- **Luminance conversion:** Each pixel’s RGB values are aggregated as $0.299R + 0.587G + 0.114B$, producing the grayscale images shown in Figures 2a and 2b.
- **CLAHE:** Contrast Limited Adaptive Histogram Equalization (Figure 2c) equalizes local histograms while limiting noise amplification, making the threshold more stable.

2.2.2. Training

Otsu’s threshold pulls a level that minimizes intra-class variance on the bimodal histogram (Figure 2d), but it can

underestimate saliva in uneven lighting. We therefore learn a bias factor that adjusts the threshold before binarization. The bias is updated as

$$t_{\text{new}} = t_{\text{otsu}} \times (1 - \text{bias}), \quad (1)$$

and we sweep this bias on the tuning set to maximize the Dice coefficient. Plausibility filters then remove regions that are either too large (non-saliva) or too small (noise) and reward blobs near the geometric center of the annotated patches. Figure 1 from Felipe’s folder summarizes the six labeled stages (a–g) that produce the cleaned mask.

Figure 2 shows the bias tuning curve that calibrates Otsu’s threshold; the selected bias of -0.18 makes the mask more permissive and closer to the annotated saliva regions (Figure 3 in the draft).

2.2.3. Testing and post-processing

With the tuned bias (Figure 2), images 9 and 10 are processed without consulting the ground truth. The same area-based filters remove unrealistic regions, but without direct knowledge of the annotated centroids, we judge centrality relative to the image center. Morphological closing (erosion followed by dilation) closes small gaps (Figure 1f), and the final mask is compared to the blind ground truth for precision/recall reporting (Figure 1f).

2.3. Method B: Supervised per-channel thresholding and Linear Discriminant Analysis (LDA)

Method B establishes a supervised baseline using the 8 tuning images and their partial ground-truth masks. The goal is to estimate pixel-level decision rules that exploit the strong fluorescence contrast of saliva under UV-C illumination. We evaluate two approaches: (i) independent per-channel thresholds and (ii) a combined linear discriminant classifier.

2.3.1. Per-channel threshold learning

For each channel $c \in \{R, G, B\}$, we compute the distribution of pixel intensities for saliva and background using the labeled tuning subset. We sweep all unique intensity values as candidate thresholds and compute the Dice coefficient [5] obtained when predicting saliva whenever $x_c \geq t$. The optimal threshold for each channel is chosen as:

$$t_c = \arg \max_t \text{Dice}(t) \quad (2)$$

A pixel is classified as saliva if any channel exceeds its optimal threshold:

$$y_{R,G,B} = 1 \text{ if } R \geq t_R \vee G \geq t_G \vee B \geq t_B \quad (3)$$

This provides a simple supervised baseline in which each channel contributes independently.

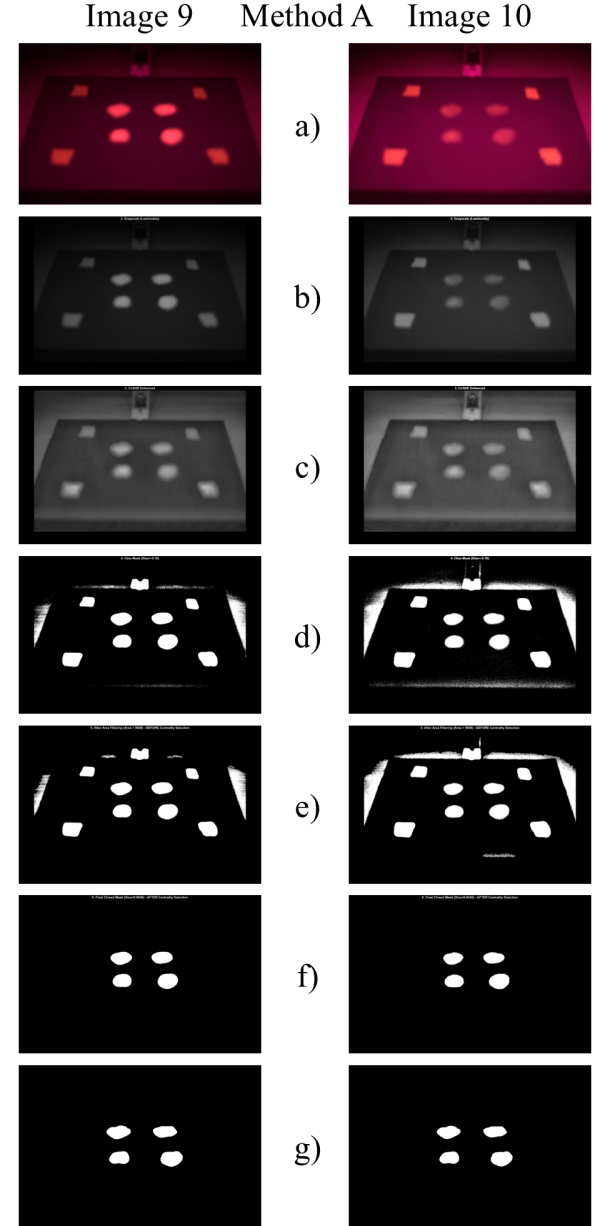


Figure 1. Method A pipeline as documented by Felipe: original UV-C image, contrast enhancement, biased thresholding, filtering, and the cleaned binary mask (panels a–g).

2.3.2. Linear Discriminant Analysis (LDA)

While independent thresholds produce axis-aligned decision boundaries, the RGB channels are strongly correlated. As shown in Figure 3, saliva pixels exhibit extremely strong positive correlations across all three channels (0.97–0.99), meaning that their colour variation is almost entirely explained by a single underlying intensity dimension. In contrast, background pixels show weak or even negative cross-

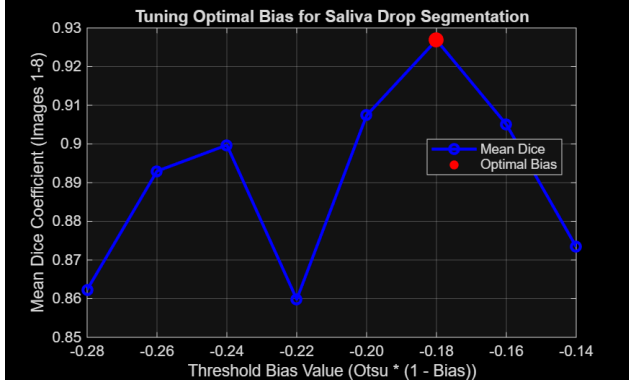


Figure 2. Bias parameter selection used to shift Otsu’s threshold toward a more permissive mask (Figure 3).

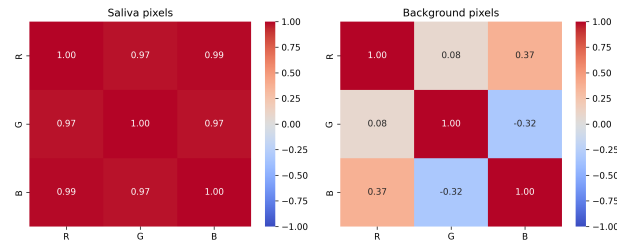


Figure 3. Correlation heatmap of RGB intensities for saliva and background pixels.

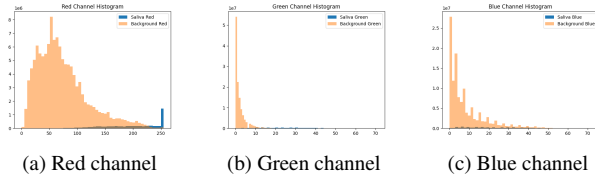


Figure 4. RGB channel histograms showing intensity distributions for saliva and background pixels.

channel correlations, indicating substantially greater chromatic variability.

The histograms illustrated in Figure 4 show the intensity distributions for each channel, demonstrating clear separation between saliva and background regions.

This difference motivates a multivariate classifier that can exploit joint RGB patterns rather than thresholding each channel independently.

We therefore learn a single discriminant direction $\mathbf{w} \in \mathbb{R}^3$ using Linear Discriminant Analysis (LDA) [7]. LDA computes a projection vector

$$\mathbf{w} = \mathbf{S}_w^{-1}(\boldsymbol{\mu}_{\text{saliva}} - \boldsymbol{\mu}_{\text{bg}}) \quad (4)$$

that maximizes the separation between saliva and background, where $\boldsymbol{\mu}_{\text{saliva}}$ and $\boldsymbol{\mu}_{\text{bg}}$ are the class means, and $\mathbf{S}_w$ is the pooled within-class covariance matrix.

Each pixel is mapped to a one-dimensional score $s = \mathbf{w}^T \mathbf{x}$.

We then sweep all unique score values on the tuning set and select the threshold T that maximizes the Dice coefficient [5].

The resulting classifier is

$$y = \begin{cases} 1 & \text{if } \mathbf{w}^T \mathbf{x} \geq T \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

This creates an oblique decision boundary in RGB space, enabling LDA to exploit correlated variations across channels and improving robustness relative to independent per-channel thresholds.

2.3.3. Pre- and Post-Processing

Before classification, a Gaussian blur is applied to suppress sensor noise while preserving the low-frequency structure of saliva fluorescence.

After thresholding, morphological closing and opening remove isolated noise pixels and fill small gaps, producing more coherent saliva regions.

2.4. Method C: exploratory k-means

Method C clusters RGB pixels with k-means. Each run uses ten random starts, and we select the solution with the smallest compactness (within-cluster sum of squares). The centroid with the highest luminance $0.299R + 0.587G + 0.114B$ is labeled as saliva, and the corresponding pixels form a binary mask that is cleaned with morphological opening and closing.

2.4.1. Pre-processing

No pre-processing beyond using raw RGB pixels was required—the UV-C contrast is already sufficient to separate saliva from the background, and Gaussian or median filters degraded performance.

2.4.2. Clustering

- Ten runs per image keep the best result by compactness.
- The brightest center identifies saliva, and its assigned pixels become the foreground mask.
- Each mask is cleaned with morphological operations (opening and closing) to remove speckles and fill holes.

2.4.3. Discussion

Experiments with $K = 2, 3, 4, 5$ show that $K = 3$ is a good precision/recall balance, $K = 4$ retains high recall while sacrificing some precision, and higher values lose too much recall. Table 1 reports the metrics for the $K = 3$ case without pre-processing; the variants that applied filtering produced lower accuracy and recall.

Table 1. Method C ($K = 3$) metrics without pre-processing.

Image	Accuracy	Precision	Recall	Dice
1.tif	0.997	0.937	0.896	0.916
2.tif	0.844	0.092	0.820	0.165
3.tif	0.990	0.813	0.949	0.876
4.tif	0.960	0.432	0.895	0.583
5.tif	0.818	0.096	0.995	0.175
6.tif	0.779	0.105	0.985	0.190
7.tif	0.987	0.671	0.923	0.777
8.tif	0.949	0.387	0.977	0.554
9.tif	0.981	0.552	0.989	0.709
10.tif	0.963	0.358	0.820	0.498
Average	0.927	0.444	0.925	0.544

2.5. Method D: probabilistic Gaussian mixture model

Method D upgrades the k-means clusters with a full Gaussian mixture model that quantifies uncertainty and chooses the most probable segmentation. Each pixel $\mathbf{x}_i = (R_i, G_i, B_i)^\top$ is modeled as

$$p(\mathbf{x}_i | \Theta) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_i | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

with $K \in \{3, 4\}$, full covariances $\boldsymbol{\Sigma}_k$, and mixture weights π_k . Expectation–Maximization is handled by `sklearn.mixture.GaussianMixture`, while our contribution is the multi-start setup, initialization, and model selection via pseudo-log-likelihood.

2.5.1. Pre-processing

- The images remain in RGB space and are reshaped into pixel arrays so the GMM learns the native color distribution.
- Candidate means $\boldsymbol{\mu}_k^{(0)}$ come from Method C’s k-means centers, helping EM start near saliva/background splits.
- No additional smoothing or filtering is applied, keeping preprocessing comparable to the other pipelines.

2.5.2. Training and model selection

- For each image and each K , we run $n_{\text{runs}} = 10$ independent GMM fits mixing k-means seeds with random starts to explore multiple basins of attraction.
- Each fit returns

$$\text{PLL}(\Theta) = \sum_{i=1}^N \log p(\mathbf{x}_i | \Theta),$$

which we use to select the most probable solution because EM only guarantees local convergence.

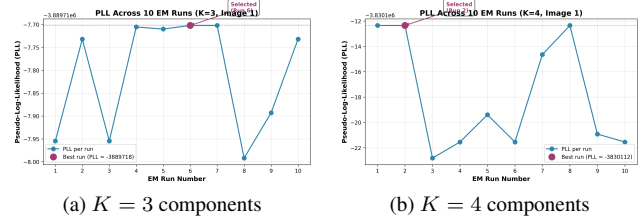


Figure 5. Pseudo-log-likelihood (PLL) across 10 EM runs for Image 1. The selected run (highlighted) maximizes PLL, demonstrating the model selection strategy.

- EM yields responsibilities

$$\gamma_{ik} = p(z_i = k | \mathbf{x}_i, \Theta^*),$$

so the foreground probability map $P_{\text{FG}}(\mathbf{x}_i)$ is given by $\gamma_{ik_{\text{fg}}}$, where k_{fg} is the component with the highest luminance $Y_k = 0.299R_k + 0.587G_k + 0.114B_k$.

Figure 5 illustrates how PLL varies across the ten EM runs for a representative image, demonstrating why selecting the highest-likelihood solution matters: different initializations converge to different local optima, and the PLL quantifies which solution best fits the RGB distribution.

2.5.3. Mask construction and evaluation

- We threshold the probability map at 0.5, yielding a soft mask that is refined with morphological opening and closing to remove noise.
- Accuracy, precision, recall, Dice, and the pseudo-log-likelihood of the selected model are recorded for every image and every K ; these per-image metrics feed the results summary and demonstrate how likelihood-guided selection aligns with segmentation quality.

This fully probabilistic workflow keeps the interpretability of k-means while adding soft assignments, likelihood-based selection, and quantified uncertainty via the posterior probability maps.

3. Results and discussion

Table 2 summarizes the average accuracy, precision, recall, Dice, and pseudo-log-likelihood (PLL) for the evaluated configurations. Method C delivers the highest precision at $K = 3$, while the GMM achieves high recall (0.897 at $K = 3$, 0.630 at $K = 4$); switching to $K = 4$ raises Dice and improves the PLL, signaling a tighter probabilistic fit to the RGB distributions.

All per-image accuracy, precision, recall, Dice, and PLL values are recorded for every run so that the aggregated numbers in Table 2 and the qualitative observations align with the logged findings. These per-image logs capture the variability across substrates and support reproducibility, allowing us to trace the numbers in the table back to the recorded output of each experiment.

Method	K	Accuracy	Precision	Recall	Dice	PLL (10^6)
Method B (LDA, supervised)	—	—	—	—	—	—
Method C (k-means, no preprocessing)	3	0.927	0.444	0.925	0.544	—
Method D (GMM)	3	0.939	0.272	0.897	0.426	-2.92
Method D (GMM)	4	0.951	0.320	0.630	0.481	-2.77

Table 2. Cross-method comparison of segmentation metrics. The GMM keeps recall high and the $K = 4$ solution improves Dice while increasing PLL relative to $K = 3$.

The probabilistic GMM absorbs bright saliva pixels that the hard clusters sometimes treat as background, yielding high recall at $K = 3$ (0.897). The addition of a fourth component, chosen by the highest PLL over ten EM runs (seeded by both k-means centers and random starts), improves precision (0.272 to 0.320) and raises Dice (0.426 to 0.481) while maintaining reasonable recall (0.630). This likelihood-guided selection, reinforced by the probability maps and morphological cleanup, ensures that each reported mask aligns with the PLL value that drove its selection.

Figure 6 showcases representative outputs from Image 1 for $K = 3$ and $K = 4$; the probability maps illustrate how the brightest Gaussian concentrates on saliva, while the thresholded masks show how morphological cleanup delivers the final binary regions.

Qualitative outputs (probability maps and binary masks) complement the numeric metrics: the posterior probabilities provide spatial confidence and the thresholded masks, once cleaned with opening/closing, reflect both the high-likelihood components and the morphological heuristics.

4. Discussion

Method B establishes a supervised baseline using per-channel thresholds and Linear Discriminant Analysis (LDA), exploiting the strong fluorescence contrast of saliva under UV-C illumination. The LDA approach creates an oblique decision boundary in RGB space, enabling it to exploit correlated variations across channels and improving robustness relative to independent per-channel thresholds. Method C offers a fast, hard-clustering baseline: $K = 3$ keeps precision around 0.44 while $K = 4$ shifts toward higher recall. Method D embraces the uncertainty of Gaussian mixtures, turning k-means centers into soft assignments while the PLL separates the best runs from the rest. The per-image metrics show that the probability map recovers gaps that hard clustering misses, achieving recall of 0.897 at $K = 3$ and 0.630 at $K = 4$, with Dice improving from 0.426 to 0.481.

Recording accuracy, precision, recall, Dice, and PLL after each fit makes the pipeline auditable, so readers can reproduce the trade-offs and confirm that the chosen components correlate with better log-likelihoods. Pending runtime and memory profiling numbers will extend the comparison

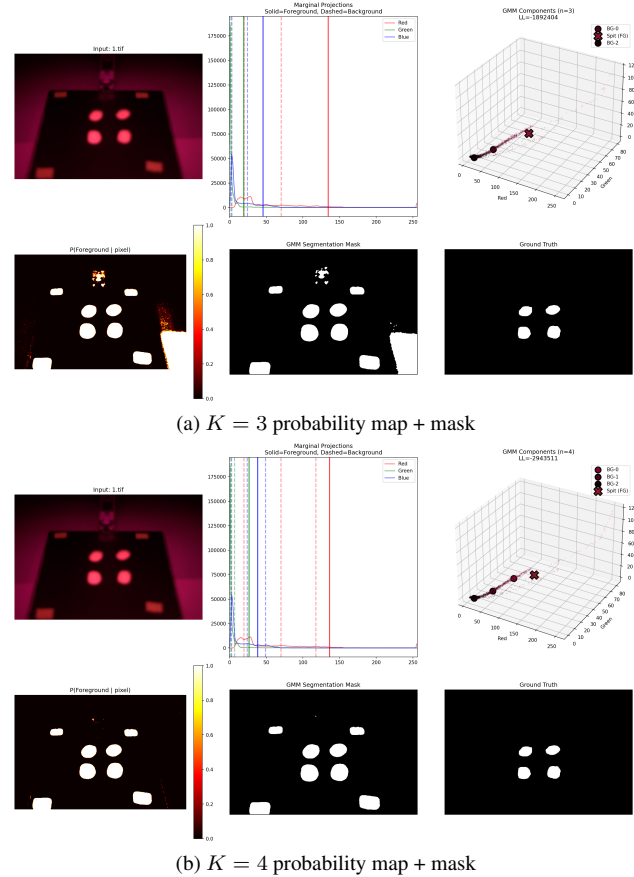


Figure 6. Gaussian mixture outputs for Image 1 at $K = 3$ and $K = 4$. The probability maps highlight the saliva component, while the cleaned masks demonstrate the final segmentation.

to computational cost, but for now the focus remains on stable segmentation in the low-data regime.

Future work will expand the dataset and explore hybrid models. Collecting more UV-C frames (or generating synthetic droplets) would help validate supervised backbones such as U-Net or transformer encoders, while additional substrates and illumination settings will test generalization. We also plan to investigate domain adaptation or attention-guided priors so the models can focus on saliva-like blobs despite new backgrounds, and to capture runtime/memory inferences on the target hardware so we can report deployment costs alongside accuracy and PLL.

5. Conclusion

This report documents a saliva segmentation workflow that combines Method A’s MATLAB thresholding, Method B’s supervised LDA classifier, Method C’s k-means clustering, and Method D’s likelihood-aware Gaussian mixture model. Method D keeps recall high while raising Dice at

$K = 4$ by selecting the most probable EM run via pseudo-log-likelihood, illustrating that probabilistic modeling and multi-start EM deliver stable masks for safety-critical contamination imagery. The detailed log of per-image metrics ensures the reported averages trace back to reproducible per-run outputs.

References

- [1] B. Allegranzi and D. Pittet. Role of hand hygiene in healthcare-associated infection prevention. *Journal of Hospital Infection*, 77(4):305–315, 2011. 1
- [2] B. Chang, R. J. Boltryk, Y. Amarouchene, and D. K. Dysthe. Aerial mucosal droplet dispersal distributions with implications for disease mitigation. *Physical Review Research*, 2(4):043391, 2020. 1, 2
- [3] N. F. de Oliveira Neto et al. The emergence of saliva as a diagnostic and prognostic tool for viral infections. *Viruses*, 16(11):1759, 2024. 1
- [4] C.-X. Deng, W. Ma, and Y. Yin. An edge detection approach of image fusion based on improved sobel operator. In *2011 4th International Congress on Image and Signal Processing*, pages 1934–1938, 2011. 2
- [5] L. R. Dice. Measures of the amount of ecologic association between species. *Ecology*, 26(3):297–302, 1945. 3, 4
- [6] A. Dwivedi, S. Joshi, R. Kumar, V. Rai, V. Chaudhary, and P. K. Mishra. Optimizing diagnosis approaches: A computational perspective for medical image processing in cholangiocarcinoma detection. *Archives of Computational Methods in Engineering*, 2025. 2
- [7] R. A. Fisher. The use of multiple measurements in taxonomic problems. *Annals of Eugenics*, 7(2):179–188, 1936. 4
- [8] H. P. Loveday, J. A. Wilson, R. J. Pratt, et al. epic3: National evidence-based guidelines for preventing healthcare-associated infections in nhs hospitals in england. *Journal of Hospital Infection*, 86(Supplement 1):S1–S70, 2014. 1
- [9] S. S. Magill, J. R. Edwards, W. Bamberg, et al. Multi-state point-prevalence survey of health care-associated infections. *New England Journal of Medicine*, 370(13):1198–1208, 2014. 1
- [10] D. Malamud. Saliva as a diagnostic fluid. *Dental Clinics of North America*, 55(1):159–178, 2011. 1
- [11] G. E. Miranda, J. M. Campos, and R. Pérez. Analysis of the fluorescence of body fluids on different surfaces for forensic applications. *Science & Justice*, 54:427–431, 2014. 1, 2
- [12] J. A. Otter, S. Yezli, J. A. G. Salkeld, and G. L. French. Evidence that contaminated surfaces contribute to the transmission of hospital pathogens and an overview of strategies to address contaminated surfaces in hospital settings. *American Journal of Infection Control*, 44(5):S6–S11, 2013. 1
- [13] T.-H. Pham and K.-D. Nguyen. Enhanced droplet analysis using generative adversarial networks. *Computer Vision and Pattern Recognition, Cornell University*, 2024. 2
- [14] T. Rahman et al. Exploring the effect of image enhancement techniques on covid-19 detection using chest x-ray images. *Computers in Biology and Medicine*, 132:104319, 2021. 2
- [15] X. Ren and S. Lai. Medical image enhancement based on laplace transform, sobel operator and histogram equalization. *Academic Journal of Computing & Information Science*, 5(6), 2022. 2
- [16] A. Sarritzu et al. droplet-ml, 2024. <https://github.com/alesarritzu/droplet-ml>. 2
- [17] S. H. Shah et al. Assessment of respiratory droplet transmission during the ophthalmic slit-lamp exam: A particle tracking analysis. *American Journal of Ophthalmology*, 222: 76–81, 2021. 1, 2
- [18] K. K.-W. To, O. T.-Y. Tsang, C. C.-Y. Yip, et al. Consistent detection of 2019 novel coronavirus in saliva. *Clinical Infectious Diseases*, 71(15):841–843, 2020. 1
- [19] D. J. Weber and W. A. Rutala. Understanding and preventing transmission of healthcare-associated pathogens due to the contaminated hospital environment. *Infection Control & Hospital Epidemiology*, 34(5):449–452, 2013. 1
- [20] F. Wu, B. Zhang, J. Wu, R. Jia, and D. Sun. Quantitative measurement and application of droplets on physical surfaces based on laser-induced fluorescence. *Transport Safety and Environment*, 6(2):tdad030, 2024. 1, 2